1 Notation

We'll investigate the generating functions for quarter-plane walks where we track how many times these walks touch the walls.

- Q is the generating function.
- Powers of variables x_i keep track of the i-th coordinate of the endpoint of a random walk. For example the monomial $x_1^3 x_2^1 x_4^7$ counts how many walks end at $(3, 1, 0, 7)$.
- Powers of variables a_i keep track of the number of times a random walk passes through a point with i-th coordinate equal to zero. For example the monomial $a_1^2 a_2$ counts how many times the random walks have passed through points with coordinates $(0, j)$ and $(i, 0)$ respectively.
- Powers of t count the length of random walks.
- Variables with overline are reciprocals like $\overline{x_1} = 1/x_1$.
- S is the Laurent polynomial generating function for the allowed step set. For example for the steps {N, E, SW} we have $S = x_2 + x_1 + \overline{x_1 x_2}$.
- We'll often use $b_i = a_1 - 1$ and $c_i = b_i/a_i = 1 - \overline{a_i}$.

Next we have some notation on extraction of coefficients

- $[d] = (1, \ldots, d)$ as a sequence.
- For a sequence $I = (i_1, \ldots, i_n)$ and a sequence $P = (p_1, \ldots, p_n)$, $x_I^P = x_{i_1}^{p_1} \ldots x_{i_n}^{p_n}$. For example $x_{[3]}^{(1,0,2)} = x_1^1 x_2^0 x_3^2 = x_1 x_3^2$. We'll usually use this notation for when we have general sequences, not in the case of the example which we would write out explicitly.
- $[u_I^P]$ stands for extracting the coefficient infront of u_I^P in the generating function expansion.
- $Q_I^J = [x_I^P]Q$ and similarly $S_I^J = [x_I^J]S$. For example if $I = (1, 3)$, $J = 0, 1$, (omiting parentheses) $Q_I^J = Q_{1,3}^{0,1}$. We will sometimes write $(Q_I^P)_J^R = Q_{I,J}^{P,R}$
- For a set I, $\mathbb{O}$ and $\mathbb{I}$ stand for sequences of $|I|$ zeros and ones respectively. For example for $I \subseteq [d]$ we might have $S_I^{-\mathbb{I}} Q_I^{\mathbb{O}}$. Analogously we will write $(Q_I^{\mathbb{I}})_J^{\mathbb{O}} = Q_{I,J}^{\mathbb{I},\mathbb{O}}$

So we have

$$Q(x_1, x_2, a_1, a_2; t) = \sum_{i,j,k} Q_{1,2}^{i,j} x_1^i x_2^j t^k$$

So far we won't extract a's because we have an unobvious choice between extracting a, b or c with various pros/cons. For example it may sometimes be easier to extract b_1^k than a_1^k but there's the added cost that the extracted term counts walks with varying powers of k. We usually won't extract with respect to t.

2 Deriving the functional equation for the 2D unlabelled random walks

We begin constructing a functional equation for Q, the generating function for the quarter-plane walks with steps from S (we'll use it for both the step set and the laurent polynomial). We use the inclusion-exclusion principle. We start with 1 (I'm not sure exactly where it comes). When we are far from the origin and the axes we can step in any of the directions from S. So we'll add tSQ because the walks will now have 1 more step and any of the walks counted by Q should be in there as well.

However, we've overdone it because the expression tSQ also counts those walks that end up below the x_1 axis or on the left of the x_2 axis. We must subtract those invalid walks. A walk will become invalid with respect to the x_1 axis if it start at coordinate $(i, 0)$ for some i, that is the x_1 axis and the next step leads it below the x_1 axis, that is with negative x_2 coordinate. The walks starting on the x_1 axis are those counted by Q_2^0. The steps in the negative x_2 directions are those counted by S_2^{-1}. The term we need to remove is $t\overline{x_2}S_2^{-1}Q_2^0$ where the t is due to the increase in step number. Analogously we need to remove the term $\overline{x_1}S_1^{-1}Q_1^0$.

However, there is a problem. We've removed too many terms, namely those that count walks on the origin and go in the SW direction, because this is the common direction counted by both the terms in the previous paragraph. Thus we need to add in the term involving walks starting at the origin (counted by $Q_{1,2}^{0,0}$) and moving in the SW direction (steps counted by $S_{1,2}^{-1,-1}$). The term is $t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$. Since this covers all cases we immediately have the functional equation

$$Q = 1 + tSQ - t\overline{x_2}S_2^{-1}Q_2^0 - t\overline{x_1}S_1^{-1}Q_1^0 + t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$$

3 Deriving the functional equation for the 2D labelled random walks

To derive the functional equation for the labelled random walks we start with the equation for the unlabelled walks and modify it to add terms counting interactions with the walls. Each of the terms will rely on inclusion-exclusion to arrive at a valid expression. First, we need to identify which terms will count walks that will have to be labelled on their step, remove those terms and add them back in with a multiplication by a_i.

We start with identifying which terms will be labelled on the next step due to an interaction with the x_1-axis. We have two cases

- Walks ending on the x_1-axis (counted by Q_2^0) moving in the positive or negative x_1 direction (steps counted by S_2^0). The term is $tS_2^0Q_2^0$.
- Walks ending one unit above the x_1-axis (counted by Q_2^1) moving in the negative x_2 direction (steps counted by S_2^{-1}). The term is $tS_2^{-1}Q_2^1$.

However, we've over-reached in this tally and we need to remove those terms that count invalid quarter-plane walks, namely those that also end up beyond the x_2 axis on the next step.

- Walks ending on the x_1-axis and the x_2-axis (counted by $Q_{1,2}^{0,0}$) travelling in the negative x_1 direction (steps counted by $S_{1,2}^{-1,0}$). The term is $t\overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}$
- Walks ending one unit above the x_1-axis and on the x_2-axis (counted by $Q_{1,2}^{0,1}$) traveling in the negative x_2 direction and negative x_1 direction. The term is $t\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1}$.

So the quarter-plane walks that need to be labelled with a_2 for every time they touch the x_1 axis are given by the term

$$t(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0})$$

We need to remove this term and add it back again with a label a_2 because the term counts those walks that will interact with the x_1 axis on their next step. Thus overall we need to add

$$t(a_2-1)(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0})$$

to the unlabelled functional equation. Analogously we need to add

$$t(a_1-1)(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0})$$

to the equation. However, we've mislabelled walks that would end at the origin on their next step. We've given them a_1 as a label in one case and a_2 as a label in the other case. Furthermore we've removed two copies of each due to the two terms. Therefore, we need to add back one of the terms, remove the terms with labels a_1 and a_2 and relabel them properly with label a_1a_2. There are three types of walks ending on the origin

- Walks ending on point $(0,1)$ (counted by $Q_{1,2}^{0,1}$) and then stepping South, i.e. step is $(0,-1)$ (counted by $S_{1,2}^{0,-1}$). The term is $tS_{1,2}^{0,-1}Q_{1,2}^{0,1}$.
- Walks ending on point $(1,0)$ (counted by $Q_{1,2}^{1,0}$) and then stepping West, i.e. step is $(-1,0)$ (counted by $S_{1,2}^{-1,0}$). The term is $tS_{1,2}^{-1,0}Q_{1,2}^{1,0}$.
- Walks ending on point $(1,1)$ (counted by $Q_{1,2}^{1,1}$) and then stepping Southwest, i.e. step is $(-1,-1)$ (counted by $S_{1,2}^{-1,-1}$). The term is $tS_{1,2}^{-1,-1}Q_{1,2}^{1,1}$.

We'll need to multiply the terms by the following

$$a_1a_2 - a_1 - a_2 + 1 = (a_1-1)(a_2-1)$$

in order to take care of the relabelling. Finally the term that we need to add to the equation is

$$t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})$$

This is our final step in the construction becuase we've taken care of all possible cases. The argument will be generalized to d dimensions in a later section. The lengthy inclusion-exclusion-type derivation yields the labelled equation

$$\begin{aligned}
Q = 1 &+ tSQ - t\overline{x_2}S_2^{-1}Q_2^0 - t\overline{x_1}S_1^{-1}Q_1^0 + t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0} \\
&+ t(a_1-1)(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0}) \\
&+ t(a_2-1)(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}) \\
&+ t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

4 Simplifying the 2D functional equation

As we saw in the previous section the functional equation for labelled 2D walks is

$$\begin{aligned}
Q = 1 &+ tSQ - t\overline{x_2}S_2^{-1}Q_2^0 - t\overline{x_1}S_1^{-1}Q_1^0 + t\overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0} \\
&+ t(a_1-1)(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0}) \\
&+ t(a_2-1)(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}) \\
&+ t(a_1-1)(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

The equation can be greatly simplified by eliminating terms that involve quarter-plane walks not ending on the axes or the origin. For that we'll first extract some coefficients.

$$\begin{aligned}
Q_1^0 &= 1 + ta_1(S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x}_2S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0}) \\
&\quad + ta_1(a_2-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}) \\
Q_2^0 &= 1 + ta_2(S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x}_1S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0}) \\
&\quad + ta_2(a_1-1)(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}) \\
Q_{1,2}^{0,0} &= 1 + ta_1a_2(S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1})
\end{aligned}$$

There is a hidden pattern in the form of the sumands in these extractions. Let

$$A = SQ - \overline{x_2}S_2^{-1}Q_2^0 - \overline{x_1}S_1^{-1}Q_1^0 + \overline{x_1x_2}S_{1,2}^{-1,-1}Q_{1,2}^{0,0}$$

Extracting coefficients yields

$$\begin{aligned}
A_1^0 &= S_1^{-1}Q_1^1 + S_1^0Q_1^0 - \overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0} - \overline{x_2}S_{1,2}^{0,-1}Q_{1,2}^{0,0} \\
A_2^0 &= S_2^{-1}Q_2^1 + S_2^0Q_2^0 - \overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1} - \overline{x_1}S_{1,2}^{-1,0}Q_{1,2}^{0,0} \\
A_{1,2}^{0,0} &= S_{1,2}^{0,-1}Q_{1,2}^{0,1} + S_{1,2}^{-1,0}Q_{1,2}^{1,0} + S_{1,2}^{-1,-1}Q_{1,2}^{1,1}
\end{aligned}$$

A quick visual inspection shows that the A terms are involved in the functional equation and the Q-extractions. This allow us to re-write the above equations in a much neater format. Using $b_i = a_1 - 1$ the equations become

$$\begin{aligned}
Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\
Q_1^0 &= 1 + ta_1A_1^0 + ta_1b_2A_{1,2}^{0,0} \\
Q_2^0 &= 1 + ta_2A_2^0 + ta_2b_1A_{1,2}^{0,0} \\
Q_{1,2}^{0,0} &= 1 + ta_1a_2A_{1,2}^{0,0}
\end{aligned}$$

Assuming the A-terms are non-zero we can back-substitute the A's in order to obtain an equation relating Q, Q_1^0, Q_2^0 and $Q_{1,2}^{0,0}$. In other words, we'll obtain a functional equation that relates Q to Q's counting walks ending on the axes or the origit. First

$$A_{1,2}^{0,0} = \frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}$$

and then

$$A_1^0 = \frac{Q_1^0 - 1}{ta_1} - (a_2 - 1)A_{1,2}^{0,0} = \frac{Q_1^0 - 1}{ta_1} - (a_2 - 1)\frac{Q_{1,2}^{0,0} - 1}{ta_1 a_2}$$
$$A_2^0 = \frac{Q_2^0 - 1}{ta_2} - (a_1 - 1)A_{1,2}^{0,0} = \frac{Q_2^0 - 1}{ta_2} - (a_1 - 1)\frac{Q_{1,2}^{0,0} - 1}{ta_1 a_2}$$

We can substitute these expressions into the initial equation along with $c_i = b_i/a_i = 1 - \overline{a_i}$

$$\begin{aligned} Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\ &= 1 + tA + tb_1\left(\frac{Q_1^0 - 1}{ta_1} - b_2\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}\right) + tb_2\left(\frac{Q_2^0 - 1}{ta_2} - b_1\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2}\right) + tb_1b_2\frac{Q_{1,2}^{0,0} - 1}{ta_1a_2} \\ &= 1 + tA + c_1(Q_1^0 - 1) + c_2(Q_2^0 - 1) - c_1c_2(Q_{1,2}^{0,0} - 1) \end{aligned}$$

Expanding the definition of A and re-arranging terms gives the following functional equation for Q relating it to Q_1^0, Q_2^0 and $Q_{1,2}^{0,0}$.

$$Q(1 - tS) - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1c_2 - t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1a_2}$$

The form is suggestive so for $I \subseteq [2]$ we let

$$K_I = c_I - t\overline{x_I}S_I^{-\mathbb{I}}$$

be the kernels with respect to S. When $I = \varnothing$ we just write $K = 1 - tS$. This allows us to re-write the above equation in the following compact way

$$KQ - K_1Q_1^0 - K_2Q_2^0 + K_{1,2}Q_{1,2}^{0,0} = \overline{a_1a_2}$$

This equation is an important milestone in further analysing the labelled quarter plane walks. Before we go on though, we need to say a few words about the cases when one or more of the A's is 0. If $A_1^0 = 0$ or $A_2^0 = 0$, this also means $A_{1,2}^{0,0} = 0$. This in turn makes the respective Q's equal to 1. For example, if only $A_{1,2}^{0,0} = 0$ then $S_{1,2}^{-1,-1}$, $S_{1,2}^{-1,0}$ and $S_{1,2}^{0,-1}$ are all zero, together with $Q_{1,2}^{0,0} = 1$.(Note: I now realize that this is not completely necessary. One could tell aprior if one or more of the Q's will be zero based on the step sets we have. We either have no positive steps or no negative steps for $A_{1,2}^{0,0}$ to be identially zero. This also suggest that we might be able to perform a different derivation that will focus on going forward rather than backward like this). Then

$$A_1^0 = \frac{Q_1^0 - 1}{ta_1}, \quad A_2^0 = \frac{Q_2^0 - 1}{ta_2}$$

So we obtain

$$\begin{aligned} Q &= 1 + tA + tb_1A_1^0 + tb_2A_2^0 + tb_1b_2A_{1,2}^{0,0} \\ &= 1 + tA + tb_1\frac{Q_1^0 - 1}{ta_1} + tb_2\frac{Q_2^0 - 1}{ta_2} \\ &= 1 + tA + c_1(Q_1^0 - 1) + c_2(Q_2^0 - 1) \end{aligned}$$

Expanding the definition of A and re-arranging terms gives the following functional equation for Q relating it to Q_1^0 and Q_2^0.

$$Q(1-tS) - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 = \overline{a_1 a_2} - c_1 c_2$$

We have a similar situation if any of the other A's are zero.

5 Generalization to d dimensions.

This section needs a major re-work since I'm not 100 percent clear on the validity of this derivation.

The notation so far has helped push the derivation of the functional equation forward. It also gives us a good idea how to generalize the building process for the functional equation to more dimensions and more general step sets. From now on we fix the numbe of dimensions to d and the step lengts to 1 in any of the coordinates. The 2D derivation used progressive coefficient extractions to signify interactions with more and more axes simultaneously. In d dimensions we have interactions with hyperplanes defined by setting a single coordinate to zero. i.e. the hyperplanes defined by x_i for i between 1 and d. We'll first derive the unlabelled version of the functional equation. We start with Q on the LHS and 1 on the RHS and progressively add/subtract terms to/from the RHS.

First we add the term extending arbitrary p-ve orthant walks (counted by Q) by one step in any of the directions (counted by a Laurent polynomial S). This gives a term tSQ but also intorduces invalid walks that reach beyond the boundary of the p-ve orthant. That is one or more of their coordinates become negative. We need to remove those terms. For coordinate $1 \leqslant i \leqslant d$ we remove those walks that step beyond the hyperplane $x_i = 0$. These are all the walks ending with coordinate $x_i = 0$ (counted by Q_i^0) stepping in the negative x_i-direction (steps counted by S_i^{-1}). The term that we need to add to the equation so far is

$$\sum_{i=1}^{d} (-1)^1 \overline{x_i} S_i^{-1} Q_i^0$$

We've removed too many walks though. To be specific we've over-removed the walks that end at points with at leats 2 coordinates equal to zero. This will be the basis step in the following induction. Assume that at step $1 \leqslant k \leqslant d$ in the process we add the term

$$\sum_{\substack{J \subseteq [d] \\ |J|=k}} (-1)^{|J|} \overline{x_J} S_J^{-\mathbb{I}} Q_J^{\mathbb{O}}$$

where we've over-compensated for the walks that end on points with at least $k+1$ coordinates equal to 0. Let J be a set of $k+1$ coordinates. We'll determine the compensating term for those walks that end at points with coordinates $x_j = 0$, $j \in J$ (some more coordinates may be zero too). These are counted by the term $Q_J^{\mathbb{O}}$. We need to compensate for steps counted in the previous step, that is, steps that go in the negative x_j direction for each $j \in J$ (steps counted by $S_J^{-\mathbb{I}}$). The term is $tS_J^{-\mathbb{I}} Q_J^{\mathbb{O}}$. Adding all these terms up yields the compensatory

term

$$\sum_{\substack{J\subseteq[d]\\|J|=k+1}} (-1)^{|J|}\overline{x_J}S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$$

where we've again over-compensated for those coordinates that end at points with at least $k+2$ coordinates, unless this is larger than d. This completes the induction. Combining all the compesatory terms into one major term gives

$$A = \sum_{J\subseteq[d]} (-1)^{|J|}\overline{x_J}S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$$

and the unlabelled problem has an equation that reads as

$$Q = 1 + tA. \tag{1}$$

We can also prove this combinatorially. All terms in A except the first term SQ deal with removing walks that end at hyperplanes and intersections of hyperplanes. Suppose we have an orthant walk w ending at a point with exactly k coordinates equal to zero. Let I be list those coordinates. All the other coordinates are non-zero, so we are allowed to step in those negative directions. Thus to make an invalid walk out of w, we need to step in at least one negative x_i direction for some $i \in I$. This type of walk is counted on the RHS in every term $S_J^{-\mathbb{I}}Q_J^{\mathbb{O}}$ where $J \cap I \neq \varnothing$. There are $|I|$ choose $|J| = k$ yada yada yada. They all add up and we get 1.

I wonder if there is a direct way to get this.

We first start with walks ending on no hyperplanes. Then we remove those that end on a hyperplane and go strictly beyond on the next step. Then we add back those that end on two hyperplanes simultaneously and go strictly beyond both. This means they have to go in direction that is negative with respect to both. Suppose that we've compensated for walks up to but not including those that end on k hyperplanes. We've overcompensated for those that end on k hyperplanes and go strictly beyond all k of them. This means that we need to compensate back for those walks that end on k hyperplanes and take a step in the negative direction with respect to all k hyperplanes. The term describing walks that end on k hyperplanes (listed in I) and go in the negative direction with respect to all k of them is $tS_I^{-\mathbb{I}}Q_I^{\mathbb{O}}$. The inclusion-exclusion process yields the formula directly.

Next we start working on the labelled version. What does A represent? Here A stands for allowed walks in the functional equation for the generating function. Indeed A contains only valid walks. Because $Q = 1 + tA$ this means that A contains all allowed walks except the empty walk. The way we build the next part is to consider which allowed walks we are going to label. First we label with a_i those that end on hyperplane $x_i = 0$ and move in an allowed direction. Those walks are counted by A_i^0. We need to multiply this term by b_i to remove the unlabelled ones and add in the labelled ones. However, we've mislabelled those allowed walks that end on two hyperplanes. Those are counted by $A_{i,j}^{0,0}$. They've been labelled by a_i, a_j and removed twice from the allowed walks, so we need to multiply the term by b_ib_j. Suppose that we've completed the process up to but not including step k. We've overcompensated for allowed walks on k hyperplanes listed in I. Those are counted by $A_I^{\mathbb{O}}$. We need to multiply this by some term involving labels to offset the wrong labels from the previous step.

Because the relabelling in the previous step for a given term $A_J^{\mathbb{O}}$ ends up with label a_J, the previous step has mislabelled the walks counted by $A_I^{\mathbb{O}}$ with labels a_J for $J \subseteq I$ and $|J| < |I|$. That is we need to multiply $A_I^{\mathbb{O}}$ with

$$a_I - \sum_{J \subseteq I, |J|<|I|} (-1)^{|I|-|J|} a_J = b_I$$

This gives the labelled equation

$$Q = 1 + t \sum_{I \subseteq [d]} b_I A_I^{\mathbb{O}} \tag{2}$$

We can write out the coefficient extraction explicitly as

$$Q = 1 + t \sum_{\substack{I,J \subseteq [d] \\ I \cap J = \varnothing}} b_I \overline{x_J} \sum_{K \subseteq I} \left(S_{J,K}^{-\mathbb{I},-\mathbb{I}} Q_{J,K}^{\mathbb{O},-\mathbb{I}} + S_{J,K}^{-\mathbb{I},\mathbb{O}} Q_{J,K}^{\mathbb{O},\mathbb{O}} \right)$$

although this doesn't seem to be necessary. Next, we perform coefficient extractions on equation (2) to obtain the following formula

$$Q_J^{\mathbb{O}} = 1 + t \sum_{I \subseteq [d]} b_I A_{I \cup J}^{\mathbb{O}} \tag{3}$$

$$= 1 + t a_J \sum_{I \subseteq [d] \setminus J} b_I A_{I,J}^{\mathbb{O},\mathbb{O}} \tag{4}$$

even though the second equation takes a little bit of work. Before we continue we have to make sure we know what to do if any of the A's are equal to zero. The following lemma characterizes this situation

Lemma 1. $A_I^{\mathbb{O}} \equiv 0$ if and only if $S_{J,I}^{-\mathbb{I},P} \equiv 0$ for all $J \subseteq [d] \setminus I$ and $P \in \{-1, 0\}^{|I|}$.

alternatively, since $S_{J_1,I}^{-\mathbb{I},P} \equiv 0 \Rightarrow S_{J_2,I}^{-\mathbb{I},P} \equiv 0$ for $J_1 \subseteq J_2$ based on simple extraction, then it only suffices to look at S_I^P Therefore we can re-write the lemma as

Lemma 2. $A_I^{\mathbb{O}} \equiv 0$ if and only if $S_I^P \equiv 0$ for all $P \in \{-1, 0\}^{|I|}$. In particular $A_I^{\mathbb{O}} \equiv 0 \Rightarrow S_I^{-\mathbb{I}} \equiv 0$.

The second part of the lemma guarantees that if an A is zero then the corresponding S will be zero in the specific term K in the master equation. The next part takes some effort but we can use backward induction on $|I|$ to show that

$$A_I^{\mathbb{O}} = \frac{(-1)^{|I|}}{t b_I} \sum_{I \subseteq J \subseteq [d]} (-1)^{|J|} c_J (Q_J^{\mathbb{O}} - 1) \tag{5}$$

Substituting these terms back into equation (2) yields the following calculation

$$Q = 1 + tA + t \sum_{\varnothing \neq I \subseteq [d]} b_I \frac{(-1)^{|I|}}{t b_I} \sum_{I \subseteq J \subseteq [d]} (-1)^{|J|} c_J (Q_J^{\mathbb{O}} - 1) \tag{6}$$

$$= 1 + tA + \sum_{\varnothing \neq I \subseteq [d]} (-1)^{|I|} \sum_{I \subseteq J \subseteq [d]} (-1)^{|J|} c_J (Q_J^{\mathbb{O}} - 1) \tag{7}$$

For each I we are taking all sets inbetween I and $[d]$ for the latter sum. How many times do we see every subset J? We see it once for every subset I. There are $2^{|J|}-1$ such subsets because $I \neq \varnothing$. Since J is non-empty, the number is odd so we have

$$Q = 1 + tA + \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|+1} c_J (Q_J^{\mathbb{O}} - 1) \tag{8}$$

$$= 1 + t \sum_{J \subseteq [d]} (-1)^{|J|} \overline{x_J} S_J^{-\mathbb{I}} Q_J^{\mathbb{O}} + \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|+1} c_J (Q_J^{\mathbb{O}} - 1) \tag{9}$$

$$= 1 + \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|} c_J + tSQ - \sum_{\varnothing \neq J \subseteq [d]} (-1)^{|J|} (c_J - t\overline{x_J} S_J^{-\mathbb{I}}) Q_J^{\mathbb{O}} \tag{10}$$

The sum of c_J's combines with 1 to create $\overline{a_{[d]}}$. Rerranging everything else to the LHS leaves us with the d-dimensional master equation

Theorem 3. *Given a step set in $d \geqslant 1$ dimensions with Laurent polynomial S, the labelled generating function satisfies the following functional equation*

$$\sum_{I \subseteq [d]} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = \overline{a_{[d]}}$$

or if we let $K_I = c_I - t\overline{x_I} S_I^{-\mathbb{I}}$ we have

$$\sum_{I \subseteq [d]} (-1)^{|I|} K_I Q_I^{\mathbb{O}} = \overline{a_{[d]}}.$$

We also have a corollary that takes care of all the cases where $S_I^{-\mathbb{I}} = 0$. We have a modified formula

Corollary 4. For everything as in the theorem we have

$$\sum_{\substack{I \subseteq [d] \\ A_I^{\mathbb{O}} \neq 0}} (-1)^{|I|} (c_I - t\overline{x_I} S_I^{-\mathbb{I}}) Q_I^{\mathbb{O}} = f(a_1, a_2, \ldots, a_d)$$

where

$$f(a_1, a_2, \ldots, a_d) = \overline{a_{[d]}} - \sum_{\substack{I \subseteq [d] \\ A_I^{\mathbb{O}} \neq 0}} (-1)^{|I|} c_I$$

which could easily be simplified further, now that I think about it.

6 Using Operator Notation to Simplify the Derivation

7 Generalizations are Handled Easily

8 Properties of the operators

9 Using the formal $\mathcal{L}^{-1}$ operator for iteration

10 Invesitgating the relationship between G, G_1, G_2 and $G_{1,2}$

11 Simple cases: Walsk never leaving the origin and walks leaving the origin but never returning

12 Simple cases - Factoring $S = T_1(x_1)T_2(x_2)$

13 Simple cases - Factoring $S = T_1(x_1) + T_2(x_2)$

14 General case of factoring $S = T_1(x_1) \dots T_d(x_d)$

15 Interesting 2D cases: factoring $S = T_1T_2$ - diagonal walks and almost diagonal walks

16 Interesting 2D cases: Full walks

17 Conjectured linear and rational relationships depending on $S_{1,2}^{-1,-1}$

18 Generalization to d dimensions and arbitrary finite step size

Using the idea in the previous section it is not difficult to derive a functional equation for unlabelled orthant walks with steps sets that contain steps of arbitrary finite size. We could say that no step goes beyond a certain distance in any coordinate. S is again the Laurent polynomial. To determine the functional equation we start with Q on the LHS and 1 on the RHS. We start by adding the term tSQ for arbitrary steps on the plane. However, we need

to remove those steps that cross beyond any hyperplane. Those are given by

$$t\sum_{j\geqslant 0}\overline{x}_i^{-j-1}S_i^{-j-1}\sum_{k\geqslant 0}^{j}Q_i^k = t(SQ)_i^{\leqslant -1}$$

The notation includes the x_i powers. Then we do two or more hyperplanes

$$t(SQ)_{i,j}^{\leqslant -1,\leqslant -1} = t(SQ)_{i,j}^{\leqslant -\mathbb{I}}$$

Finally the generating function satisfies

$$Q = 1 + tA$$

where

$$A = \sum_{I\subseteq[d]}(-1)^{|I|}(SQ)_I^{\leqslant -\mathbb{I}}$$

is the allowed walks on the RHS. Labelling is straight-forward from here on because we already have the set of allowed walks. Extracting the appropriate coefficients from this yields the parts that need to be labelled. How can we be sure? the SQ term involves all walks, allowed or not. The rest of the terms clean up the terms that are not allowed. Extracting $[x_I^{\mathbb{O}}]$ from SQ yeilds those walks that end up on the intersection of the I hyperplanes after we've built the new step. Thus we have the same situation as before

$$Q = 1 + t\sum_{I\subseteq[d]} b_I A_I^{\mathbb{O}}$$

Consequently the same analysis follows as in the previous section, considering all the A's are non-zero. We arrive at the same functional equation

$$\begin{aligned}
Q &= 1 + tA + \sum_{\varnothing\neq J\subseteq[d]}(-1)^{|J|+1}c_J(Q_J^{\mathbb{O}} - 1) && (11)\\
&= 1 + t\sum_{J\subseteq[d]}(-1)^{|J|}(SQ)_J^{\leqslant -\mathbb{I}} + \sum_{\varnothing\neq J\subseteq[d]}(-1)^{|J|+1}c_J(Q_J^{\mathbb{O}} - 1) && (12)\\
&= tSQ + 1 + \sum_{\varnothing\neq J\subseteq[d]}(-1)^{|J|}c_J - \sum_{\varnothing\neq J\subseteq[d]}(-1)^{|J|}(c_J Q_J^{\mathbb{O}} - t(SQ)_J^{\leqslant -\mathbb{I}}) && (13)\\
&= tSQ + \overline{a_1 a_2} - \sum_{\varnothing\neq J\subseteq[d]}(-1)^{|J|}(c_J Q_J^{\mathbb{O}} - t(SQ)_J^{\leqslant -\mathbb{I}}) && (14)\\
& && (15)
\end{aligned}$$

So we have

$$\begin{aligned}
&(1 - tS)Q + \sum_{\varnothing\neq J\subseteq[d]}(-1)^{|J|}(c_J Q_J^{\mathbb{O}} - t(SQ)_J^{\leqslant -\mathbb{I}}) = \overline{a_1 a_2} && (16)\\
& && (17)
\end{aligned}$$

as our functional equation. I've definitely not checked it for validity but oh well. I think it's correct. The trouble is that we haven't been able to quite remove all the Q terms due to

steps going 2 or more units negatively in any of the possible directions. For example if we have a full set of d-dimensional walks of steps of size at most 2, there are $2^{|J|}-1$ new Q terms in each term of the sum. This gives a much larger total of 3^d-2^d new terms inside the equation. That's all for now folks. This little excursion into the d-dimensional version of things gave me ideas on how to prove the easier things in a less-convoluted way.

19 Further analysis of the 2D equation

We can perform coefficient extractions of the master equation to get the following working forms

$$(1-tS)Q-(c_1-t\overline{x_1}S_1^{-1})Q_1^0-(c_2-t\overline{x_2}S_2^{-1})Q_2^0+(c_1c_2-t\overline{x_1x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0}=\overline{a_1a_2} \quad (18)$$

$$(1-ta_1S_1^0)Q_1^0-(c_2-ta_1\overline{x_2}S_{1,2}^{0,-1})Q_{1,2}^{0,0}-ta_1(S_1^{-1}Q_1^1-\overline{x_2}S_{1,2}^{-1,-1}Q_{1,2}^{1,0})=\overline{a_2} \quad (19)$$

$$(1-ta_2S_2^0)Q_2^0-(c_1-ta_2\overline{x_1}S_{1,2}^{-1,0})Q_{1,2}^{0,0}-ta_2(S_2^{-1}Q_2^1-\overline{x_1}S_{1,2}^{-1,-1}Q_{1,2}^{0,1})=\overline{a_1} \quad (20)$$

$$Q_{1,2}^{0,0}-ta_1a_2(S_{1,2}^{-1,-1}Q_{1,2}^{1,1}+S_{1,2}^{-1,0}Q_{1,2}^{1,0}+S_{1,2}^{0,-1}Q_{1,2}^{0,-1})=1 \quad (21)$$

There might be some operator form that will allow us to write these out exactly.

20 Diagonal steps: Deriving the Extraction-Ready Functional Equations

Let's take the case of the diagonal steps with a finite group G of order 4. Let $\bullet : x_1 \mapsto \overline{x_1}$ and $* : x_2 \mapsto \overline{x_2}$. We have

$$K^\bullet = K^* = K, \quad K_2^\bullet = K_2, \quad K_1^* = K_1$$

while the others change. We can also solve when $K=0$ for x_1. There are two roots

$$\sigma_1 = \frac{1-\sqrt{1-4t^2(S_1^{-1})^2}}{2tS_1^{-1}}, \quad \overline{\sigma_1} = \frac{1+\sqrt{1-4t^2(S_1^{-1})^2}}{2tS_1^{-1}}$$

with the extra property that

$$\sigma_1+\overline{\sigma_1} = \frac{1}{tS_1^{-1}}.$$

We can take the master equation and apply $*$ to it

$$-KQ+K_1Q_1^0+K_2Q_2^0-K_{1,2}Q_{1,2}^{0,0}+\alpha=0$$
$$-KQ^*+K_1Q_1^{0,*}+K_2^*Q_2^0-K_{1,2}^*Q_{1,2}^{0,0}+\alpha=0$$

From here on we will input $x_1=\sigma_1$ to rid ourselves of the kernel K. There will be an overloading of notation

$$K_1Q_1^0+K_2Q_2^0-K_{1,2}Q_{1,2}^{0,0}+\alpha=0$$
$$K_1Q_1^{0,*}+K_2^*Q_2^0-K_{1,2}^*Q_{1,2}^{0,0}+\alpha=0$$

We can multiply both equation by something to clear Q_2^0 because it involves σ_1 while the other term involves only x_2. We have

$$K_1K_2^*Q_1^0 + K_2^*K_2Q_2^0 - K_2^*K_{1,2}Q_{1,2}^{0,0} + \alpha K_2^* = 0$$
$$K_1K_2Q_1^{0,*} + K_2K_2^*Q_2^0 - K_2K_{1,2}^*Q_{1,2}^{0,0} + \alpha K_2 = 0$$

We subtract these two equations to obtain

$$K_1(K_2^*Q_1 - K_2Q_1^*) - (K_2^*K_{1,2} - K_2K_{1,2}^*)Q_{1,2} + \alpha(K_2^* - K_2) = 0$$
$$K_2^*Q_1 - K_2Q_1^* - \frac{K_2^*K_{1,2} - K_2K_{1,2}^*}{K_1}Q_{1,2} + \alpha\frac{K_2^* - K_2}{K_1} = 0$$

First we have

$$\begin{aligned} K_2^* - K_2 &= t(\overline{x_2} - x_2)S_2^{-1}(\sigma_1) \\ &= t(\overline{x_2} - x_2)(\overline{\sigma_1} + \sigma_1) \\ &= \frac{\overline{x_2} - x_2}{\overline{x_2} + x_2} \end{aligned}$$

We have

$$\begin{aligned} K_2^*K_{1,2} - K_2K_{1,2}^* &= (c_2 - tx_2S_2^{-1})\,(c_1c_2 - t\overline{\sigma_1 x_2}) - (c_2 - t\overline{x_2}S_2^{-1})\,(c_1c_2 - t\overline{\sigma_1}x_2) \\ &= c_1c_2t(\overline{x_2} - x_2)S_2^{-1} - c_2t\overline{\sigma_1}(\overline{x_2} - x_2) \\ &= c_2t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) \end{aligned}$$

If we let $v = \overline{x_2} + x_2$ we find

$$\sigma_1 = \frac{1 - \sqrt{1 - 4t^2v^2}}{2tv}, \quad \overline{\sigma_1} = \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv}$$

We have

$$\begin{aligned} (c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1}) &= \frac{c1}{tv} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \\ &= \frac{1}{\overline{x_2} + x_2}\left(-\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2}\right) \end{aligned}$$

Also

$$\begin{aligned} K_1 &= c_1 - tv\overline{\sigma_1} \\ &= c_1 - tv\frac{1 + \sqrt{1 - 4t^2v^2}}{2tv} \\ &= 1 - \overline{a_1} - \frac{1 + \sqrt{1 - 4t^2v^2}}{2} \\ &= -\overline{a_1} + \frac{1 - \sqrt{1 - 4t^2v^2}}{2} \end{aligned}$$

Thus we finally have

$$\begin{aligned}\frac{1}{K_1}(K_2^* K_{1,2} - K_2 K_{1,2}^*) &= -\frac{c_2 t(\overline{x_2} - x_2)(c_1(\sigma_1 + \overline{\sigma_1}) - \overline{\sigma_1})}{c_1 - t\overline{\sigma_1}v} \\ &= -\frac{c_2 t(\overline{x_2} - x_2)(-1)}{tv} \\ &= \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2}\end{aligned}$$

Thus the equation will now read as

$$K_2^* Q_1 - K_2 Q_1^* - \frac{c_2(\overline{x_2} - x_2)}{\overline{x_2} + x_2} Q_{1,2} + \alpha \frac{\overline{x_2} - x_2}{(\overline{x_2} + x_2)K_1} = 0$$

We just need to take care of the remaining terms. We'll do this by multiplying by $\overline{x_2} + x_2$. We have

$$\begin{aligned}R^* &= (\overline{x_2} + x_2)K_2^* = c_2(\overline{x_2} + x_2) - x_2 = c_2\overline{x_2} - \overline{a_2}x_2 \\ R &= (\overline{x_2} + x_2)K_2 = c_2(\overline{x_2} + x_2) - \overline{x_2} = c_2 x_2 - \overline{a_2 x_2}\end{aligned}$$

We have a couple of relations here

$$\begin{aligned}R^* - R &= v(K_2^* - K_2) = \overline{x_2} - x_2 \\ \overline{x_2}R^* - x_2 R &= \overline{x_2}c_2 v - 1 - x_2 c_2 v + 1 \\ &= c_2 v(\overline{x_2} - x_2)\end{aligned}$$

The equation becomes

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^0(x_2) + (\overline{a_2 x_2} - c_2 x_2)Q_1^0(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + (\overline{x_2} - x_2)\overline{K}_1\alpha = 0$$

and it's ready for extraction. Let $\delta_0 = \overline{a_1}(\overline{x_2} - x_2)\overline{K}_1$. We re-write the equation as

$$R^* Q_1^0 - R Q_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 = 0$$

We'll use the master equation

$$(1 - tS)Q - (c_1 - t\overline{x_1}S_1^{-1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2}S_{1,2}^{-1,-1})Q_{1,2}^{0,0} = \overline{a_1 a_2}$$

to further our work here. For diagonal steps it is

$$(1 - tS)Q - (c_1 - tv\overline{x_1})Q_1^0 - (c_2 - t\overline{x_2}S_2^{-1})Q_2^0 + (c_1 c_2 - t\overline{x_1 x_2})Q_{1,2}^{0,0} = \overline{a_1 a_2}$$

extracting the $[x_1^0]$ coefficient yields

$$Q_1^0 = \overline{a_2} + c_2 Q_{1,2}^{0,0} + a_1 tv Q_1^1$$

We can extract higher coefficients too. Let $k \geqslant 1$. Then the $[x_2^k]$ coefficient is

$$\begin{aligned}0 &= Q_1^k - tS_1^{-1}Q_1^{k+1} - tS_1^0Q_1^k - tS_1^1Q_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(S_{1,2}^{-1,-1}Q_{1,2}^{k+1,0} + S_{1,2}^{0,-1}Q_{1,2}^{k,0} + S_{1,2}^{1,-1}Q_{1,2}^{k-1,0}) \\ &= Q_1^k - tvQ_1^{k+1} - tvQ_1^{k-1} - c_2Q_{1,2}^{k,0} + t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})\end{aligned}$$

We can re-write the last one in a form more convenient for the next few steps

$$Q_1^k = tv(Q_1^{k+1} + Q_1^{k-1}) + c_2Q_{1,2}^{k,0} - t\overline{x_2}(Q_{1,2}^{k+1,0} + Q_{1,2}^{k-1,0})$$

We'll use induction to prove a nice formula but for that we'll first need to establish another base case because of this splitting of cases for the coefficient extraction. We have

$$\begin{aligned}Q_1^0 &= \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1 \\ Q_1^{0,*} &= \overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}\end{aligned}$$

Combining them with equation (20) gives

$$\begin{aligned}0 &= R^*Q_1^0 - RQ_1^{0,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= R^*(\overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^1) - R(\overline{a_2} + c_2Q_{1,2}^{0,0} + a_1tvQ_1^{1,*}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + (R^* - R)(\overline{a_2} + c_2Q_{1,2}^{0,0}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{0,0} + \overline{a_2}\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a_2}(\overline{x_2} - x_2) + \overline{a_2}\delta_0 \\ &= a_1tv(R^*Q_1^1 - RQ_1^{1,*}) + \overline{a_2}((\overline{x_2} - x_2) + \delta_0)\end{aligned}$$

So we can write the equation

$$R^*Q_1^1 - RQ_1^{1,*} + \overline{a_2}\delta_1 = 0$$

where

$$\delta_1 = \frac{(\overline{x_2} - x_2) + \delta_0}{a_1tv}$$

This equation together with equation (20) form the base cases for the following lemma

Lemma 5. Let $m \geqslant 0$ be an integer. Then

$$R^*Q_1^m - RQ_1^{m,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0 \tag{22}$$

where δ_m is not a function of a_2.

Proof. We'll show this by induction. We have the base cases $m = 0$ and $m = 1$. The latter one fits the pattern because $Q_{1,2}^{1,0} = 0$ since we cannot have any walks end on coordinates $(1, 0)$. Now we assume that the formula is true for all cases up to and including m. Then

we use formula (20)

$$\begin{aligned}
&R^*Q_1^m - RQ_1^{m,*}\\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(R^*Q_1^{m-1} - RQ_1^{m-1,*})\\
&+ c_2(R^* - R)Q_{1,2}^{m,0} - t(\overline{x_2}R^* - x_2R)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0})\\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + tv(c_2(\overline{x_2} - x_2)Q_{1,2}^{m-1,0} - \overline{a_2}\delta_{m-1})\\
&+ c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)(Q_{1,2}^{m+1,0} + Q_{1,2}^{m-1,0})\\
=&tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) + c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0}) - \overline{a_2}tv\delta_{m-1})
\end{aligned}$$

Combining this with equation (22) leads to

$$\begin{aligned}
0 &= tv(R^*Q_1^{m+1} - RQ_1^{m+1,*}) - c_2tv(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} - \overline{a_2}tv\delta_{m-1} + \overline{a_2}\delta_m\\
0 &= R^*Q_1^{m+1} - RQ_1^{m+1,*} - c_2(\overline{x_2} - x_2)Q_{1,2}^{m+1,0} + \overline{a_2}\delta_{m+1}
\end{aligned}$$

where

$$\delta_{m+1} = \frac{\delta_m}{tv} - \delta_{m-1}$$

This function δ_{m+1} is also not dependent on a_2 and this completes the proof! □

There is an equivalent statement for Q_2^n. We can expand R^* and R to a form we can use

$$(c_2\overline{x_2} - \overline{a_2}x_2)Q_1^m(x_2) - (c_2x_2 - \overline{a_2x_2})Q_1^m(\overline{x_2}) - c_2(\overline{x_2} - x_2)Q_{1,2}^{m,0} + \overline{a_2}\delta_m = 0$$

We start by extracting the $[x_2^1]$ coefficient

$$\begin{aligned}
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} - c_2Q_{1,2}^{m,0} + c_2Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0\\
&c_2Q_{1,2}^{m,2} - \overline{a_2}Q_{1,2}^{m,0} + [x_2^1]\overline{a_2}\delta_m = 0
\end{aligned}$$

We can also extract the $[x_2^{n+1}]$ coefficient for $n \geqslant 2$

$$c_2Q_{1,2}^{m,n+2} - \overline{a_2}Q_{1,2}^{m,n} + [x_2^{n+1}]\overline{a_2}\delta_m = 0$$

This formula also contains the $n = 0$ case so we treat them together. We can set $a_2 = 1$ to get

$$-\breve{Q}_{1,2}^{m,n} + [x_2^{n+1}]\delta_m = 0$$

Substituting this back into the above equation gives

$$b_2Q_{1,2}^{m,n+2} = Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}$$

An equivalent formula can be derived if the process is followed through the use of x_1 instead of x_2 in the analysis.

Corollary 6. For integers $m, n \geqslant 0$ we have the following relations

$$\begin{aligned}
b_1Q_{1,2}^{m+2,n} &= Q_{1,2}^{m,n} - \widehat{Q}_{1,2}^{m,n}\\
b_2Q_{1,2}^{m,n+2} &= Q_{1,2}^{m,n} - \breve{Q}_{1,2}^{m,n}
\end{aligned}$$

We can sum up those equations to obtain functional equations

Corollary 7. For integers $m \geqslant 0$ we have the following relations

$$\begin{aligned}
Q_1^m - \widehat{Q}_1^m &= b_1 Q_1^{m+2} \\
\breve{Q}_1^m - \breve{\widehat{Q}}_1^m &= b_1 \breve{Q}_1^{m+2} \\
x_1^2(Q_2^n - \widehat{Q}_2^n) &= b_1(Q_2^n - Q_{1,2}^{1,n} x_1 - Q_{1,2}^{0,n}) \\
x_1^2(\breve{Q}_2^n - \breve{\widehat{Q}}_2^n) &= b_1(\breve{Q}_2^n - \breve{Q}_{1,2}^{1,n} x_1 - \breve{Q}_{1,2}^{0,n}) \\
x_1^2(Q - \widehat{Q}) &= b_1(Q - x_1 Q_1^1 - Q_1^0) \\
x_1^2(\breve{Q} - \breve{\widehat{Q}}) &= b_1(\breve{Q} - x_1 \breve{Q}_1^1 - \breve{Q}_1^0)
\end{aligned}$$

Corollary 8. For integers $n \geqslant 0$ we have the following relations

$$\begin{aligned}
Q_2^n - \breve{Q}_2^n &= b_2 Q_2^{n+2} \\
\widehat{Q}_2^n - \breve{\widehat{Q}}_2^n &= b_2 \widehat{Q}_2^{n+2} \\
x_2^2(Q_1^m - \breve{Q}_1^m) &= b_2(Q_1^m - Q_{1,2}^{m,1} x_1 - Q_{1,2}^{m,0}) \\
x_2^2(\widehat{Q}_1^m - \breve{\widehat{Q}}_1^m) &= b_2(\widehat{Q}_1^m - \widehat{Q}_{1,2}^{m,1} x_1 - \widehat{Q}_{1,2}^{m,0}) \\
x_2^2(Q - \breve{Q}) &= b_2(Q - x_2 Q_2^1 - Q_2^0) \\
x_2^2(\widehat{Q} - \breve{\widehat{Q}}) &= b_2(\widehat{Q} - x_2 \widehat{Q}_2^1 - \widehat{Q}_2^0)
\end{aligned}$$

Every other equation is obtained from the previous one by either setting $a_1 = 1$ or $a_2 = 1$. Here we can hope to set some telescoping sum type of thing in order to say more about $Q_{1,2}^{0,0}$

Now we can move onto the main result about the diagonal walks and it is that the function for labelled walks ending at $(0, 0)$ is connected to the functions for semi-labelled and unlabelled walks endind at $(0, 0)$. We can use the master equation to extract the $[x_1^0 x_2^0]$ and $[x_1^1 x_2^1]$ coefficients yielding the following two relations

$$\begin{aligned}
Q_{1,2}^{0,0} &= 1 + t a_1 a_2 Q_{1,2}^{1,1} \\
Q_{1,2}^{1,1} &= t(Q_{1,2}^{0,0} + Q_{1,2}^{2,0} + Q_{1,2}^{0,2} + Q_{1,2}^{2,2})
\end{aligned}$$

We also have

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,2} &= b_1(Q_{1,2}^{2,0} - \breve{Q}_{1,2}^{2,0}) \\
&= b_1 Q_{1,2}^{2,0} - b_1 \breve{Q}_{1,2}^{2,0} \\
&= Q_{1,2}^{0,0} - \widehat{Q}_{1,2}^{0,0} - \breve{Q}_{1,2}^{0,0} + \breve{\widehat{Q}}_{1,2}^{0,0}
\end{aligned}$$

together with

$$\begin{aligned}
b_1 b_2 Q_{1,2}^{2,0} &= b_2 Q_{1,2}^{0,0} - b_2 \widehat{Q}_{1,2}^{0,0} \\
b_1 b_2 Q_{1,2}^{0,2} &= b_1 Q_{1,2}^{0,0} - b_1 \breve{Q}_{1,2}^{0,0}
\end{aligned}$$

We we have

$$\begin{aligned} b_1b_2Q^{1,1}_{1,2} &= t(b_1b_2Q^{0,0}_{1,2} + b_1b_2Q^{2,0}_{1,2} + b_1b_2Q^{0,2}_{1,2} + b_1b_2Q^{2,2}_{1,2}) \\ &= t(b_1b_2Q^{0,0}_{1,2} + b_2Q^{0,0}_{1,2} - b_2\widehat{Q}^{0,0}_{1,2} + b_1Q^{0,0}_{1,2} - b_1\breve{Q}^{0,0}_{1,2} + Q^{0,0}_{1,2} - \widehat{Q}^{0,0}_{1,2} - \breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \\ &= t((b_1b_2 + b_1 + b_2 + 1)Q^{0,0}_{1,2} - (b_2+1)\widehat{Q}^{0,0}_{1,2} - (b_1+1)\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \\ &= t(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \end{aligned}$$

We can finally write

$$\begin{aligned} b_1b_2Q^{0,0}_{1,2} &= b_1b_2 + ta_1a_2b_1b_2Q^{1,1}_{1,2} \\ &= b_1b_2 + t^2a_1a_2(a_1a_2Q^{0,0}_{1,2} - a_2\widehat{Q}^{0,0}_{1,2} - a_1\breve{Q}^{0,0}_{1,2} + \breve{\widehat{Q}}^{0,0}_{1,2}) \end{aligned}$$

Let $G(b_1, b_2)$ stand for $Q(0, 0, b_1, b_2)$ and G_i has $b_i = 0$. Then we can write it all as

$$b_1b_2G = b_1b_2 + t^2a_1a_2(a_1a_2G - a_2G_1 - a_1G_2 + G_{1,2})$$

or written as a linear combination

$$(b_1b_2 - t^2a_1^2a_2^2)G + t^2a_1a_2^2G_1 + t^2a_1^2a_2G_2 - t^2a_1a_2G_{1,2} = b_1b_2$$

We can divide out by a_1a_2 to have the nicer expression

$$(c_1c_2 - a_1a_2t^2)G + a_2t^2G_1 + a_1t^2G_2 - t^2G_{1,2} = c_1c_2$$

We can even write

$$G = \frac{(c_1c_2 - t^2(a_2G_1 + a_1G_2 - G_{1,2})}{c_1c_2 - a_1a_2t^2}$$